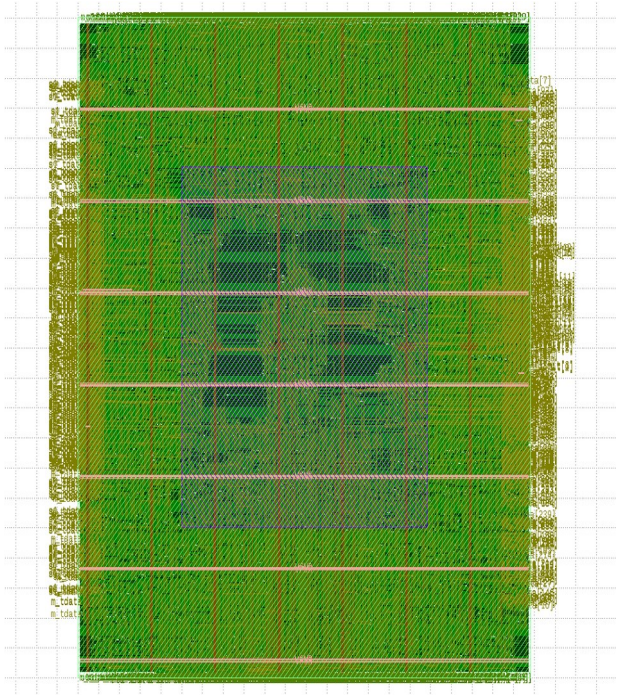
SARCATHON 2026 · SEMICON HACKATHON

AXI4-Stream to PCIe 6.0 Bridge

RTL integration, Vivado optimisation and SKY130 implementation

Team RoboKnights

Aditya Jain · Swaray Agrawal · Kshitij Seshadri



SKY130 interconnect layout view

24 October 2026

FPGA: Xilinx xcu250 | ASIC: SkyWater SKY130A

Round 1 — RTL development and initial verification

By 15 October, the three layers passed their unit tests and were connected in a first system-level loopback.

Interconnect

Aditya

- Dual 512-bit AXI4-Stream ingest ports (SmartNIC and ML).
- Weighted arbitration, session / offset tracking and async FIFO.
- Four RTL bugs fixed; 128/128 checked beats, zero errors.

Link layer

Swaray

- Write and replay buffers, 256-byte Flit framing, DLLP parsing.
- CRC/FEC and TLP-related blocks individually unit-tested.
- Transaction blocks not yet wired into the complete path.

Protocol / PHY

Kshitij

- Gray/PAM4 lane mapping and logical PHY loopback.
- Reset, configuration, MSI-X and sideband controls.
- 4/4 initial cocotb tests passed; lane proofs checked.

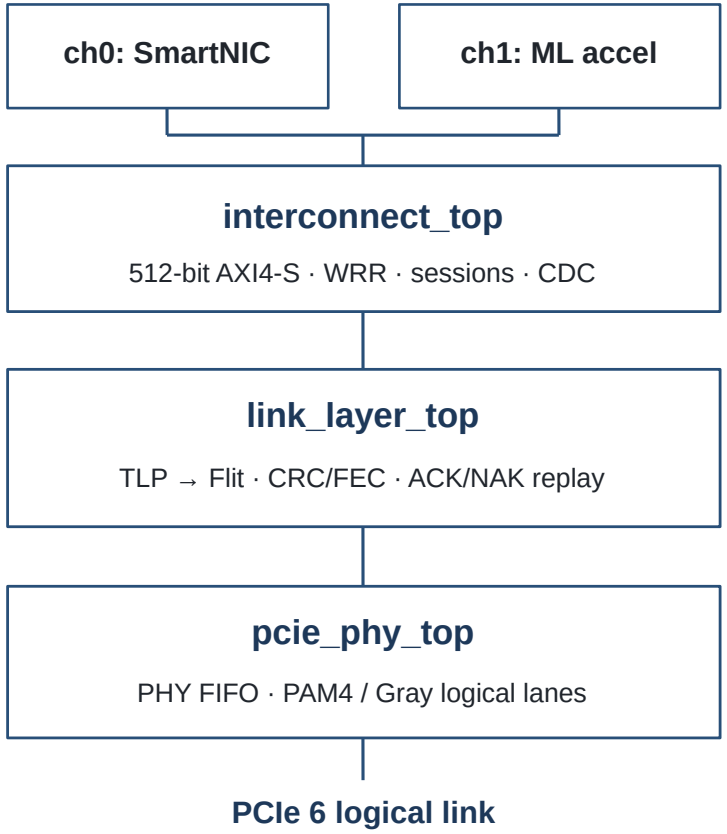
Status at submission

Connectivity was demonstrated; end-to-end TLP data correctness, complete address handling and link-to-PHY backpressure remained open.

Round 2 — full integration and regression testing

Closed the gaps between the three RTL layers before committing to FPGA performance optimisation.

Simplified data path



What changed

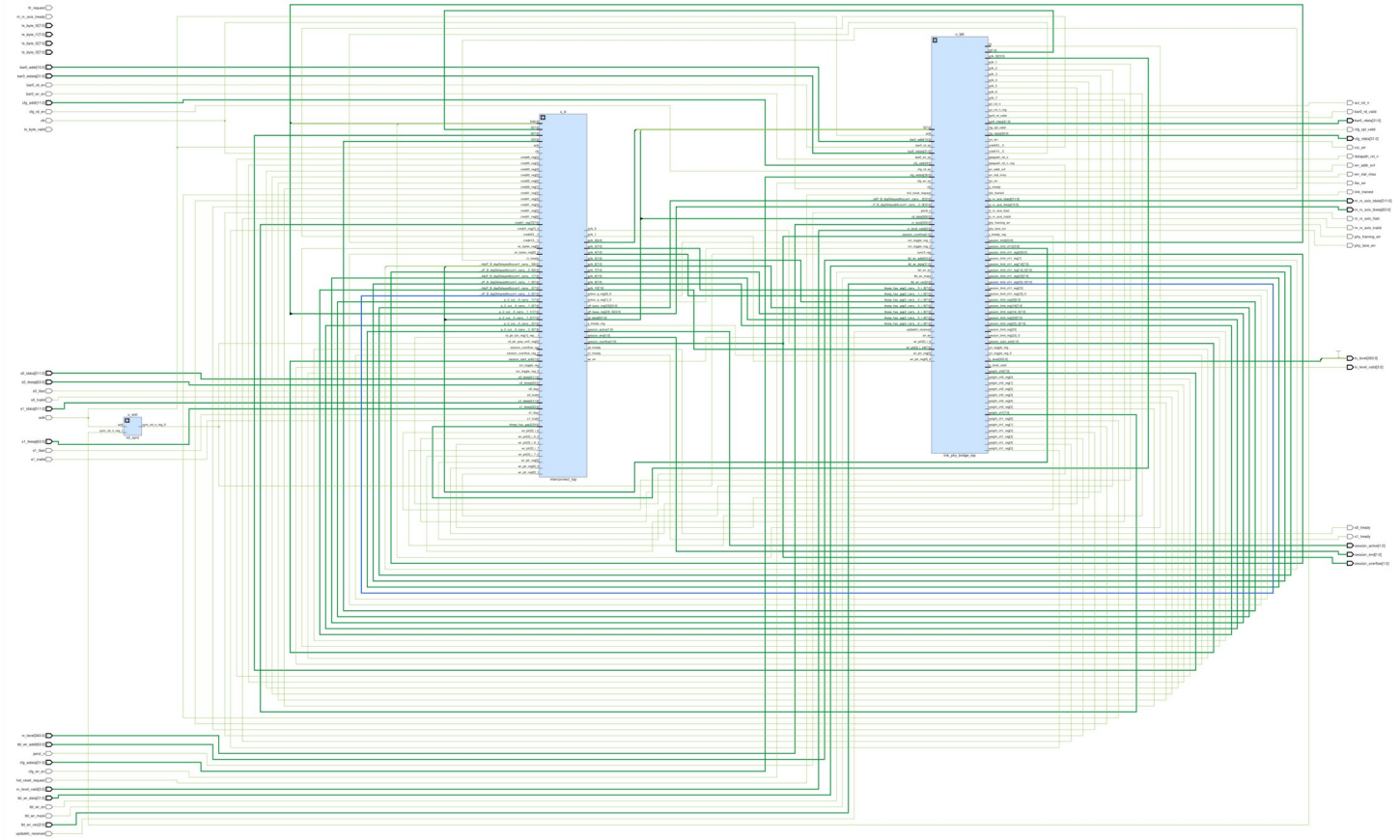
- Link-to-PHY FIFO and explicit ready / backpressure.
- Shared TUSER definition; host BAR0 registers and address translation.
- TLP packetisation, ACK/NAK replay and link-down recovery.
- Corrected malformed TLP, overflow, CRC/FEC and cross-layer data-loss cases.

Verification at the Round 2 cutoff

144/144	TLPs delivered in Step 1 integration test
73/73	Required coverage bins across 40 random seeds
4/4	Formal proofs completed
46	Protected-code regression guards

Vivado synthesis — generated bridge schematic

Structural connectivity view from Vivado for bridge_top; shown as synthesis evidence, not a timing or post-route result.



Source: synth_design_schematic(1).pdf · Vivado Nlview, bridge_top · October 24, 2026

How to read the figure

Hierarchy

The synthesized bridge_top view contains interconnect_top, link_phy_bridge_top and reset/control structures.

Visible interfaces

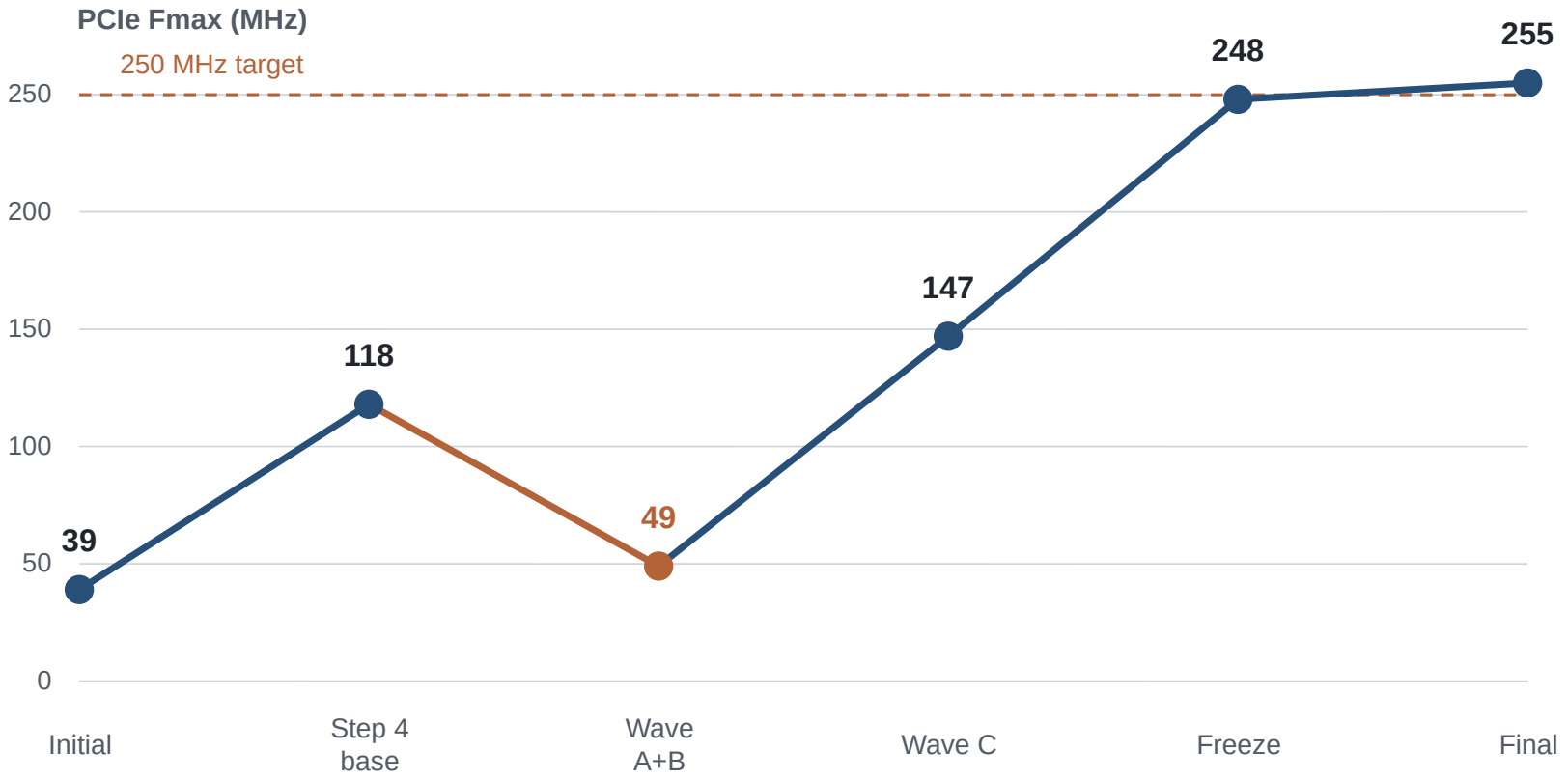
AXI4-Stream data/keep/valid-ready, BAR0/config, link status, reset and PHY lane signals.

Interpretation

Generated netlist connectivity; functional and timing claims are supported separately by testbench and Vivado reports.

Vivado — timing progression across the optimisation runs

PCIe clock target: 250 MHz (4.0 ns) on xcu250, Vivado 2026.1.



19,046 → 0 failing endpoints (Step 4 base → final)

What the timing reports showed

118 → 49 MHz

Widening the datapath created an unrolled 51-level CRC-64 critical path.

Recovery

Split CRC computation, constant-mask XOR trees, pipelined RX processing and decoupled TREADY logic.

RTL changes that mattered most

Widen the datapath for throughput, pipeline it for clock speed, then fix the protocol so no data is lost.

Bottleneck	RTL change	Measured result
32-bit TX serialiser, one TLP per Flit	512-bit tlp_tx_wide engine; merges beats into Max-Payload TLPs, split at 4 KB	**ML 0.06 → 6.17 GB/s** about 100×
RX decoded 1 DW per clock; PHY 32× slower than link	16-DW/clock RX delimiter; wide PHY (8 B/lane/clk) with FIFO back-pressure	**NIC p99 58 → 0.59 µs** 4.84 Mpps SmartNIC IMIX
CRC-64/FEC: 51 logic levels in one cycle	Pipelined check; CRC/FEC as constant-mask XOR trees	**49 → 248 MHz**
TKEEP → popcount → limit → TREADY: 27 ns	TREADY independent of TKEEP; near-limit beat checked one cycle later (X.1)	**255 MHz, all paths met** ASIC slack −12 → −0.5 ns
Replay never ACKed, FEC couldn't correct, link hung on cable pull	ACK/NAK replay, SEC-DED FEC + CRC-64, credit flow control, link recovery (K6)	**Zero data loss** 1,069 errors corrected, 7 cable pulls recovered, 40 random seeds

Overall 118 → 255 MHz (2.2×) with 2% more LUTs, because the wide path replaced the serial one rather than adding to it.

Final FPGA results and workload measurements

FPGA specifications are from the final X.1–X.4 ledger; workload results come from RTL simulation.

Final FPGA implementation (xcu250)

Parameter	Step 4 base	Final
PCIe Fmax	118 MHz	255 MHz
AXI Fmax	—	221 MHz
WNS @ 4 ns	−4.439 ns	+0.072 ns
CLB LUTs	56,043	57,245
Flip-flops	24,482	39,269
Failing endpoints	19,046	0

Traffic results (simulation only)

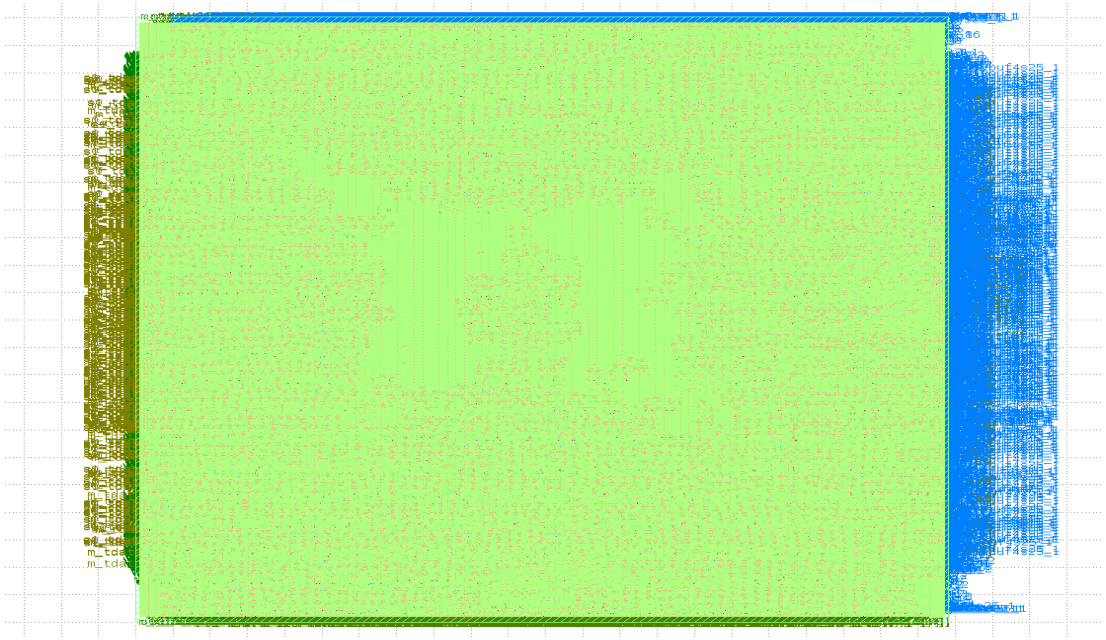
Workload / metric	Base	Optimized
ML tensor throughput	0.061 GB/s	6.17 GB/s
SmartNIC IMIX rate	0.15 Mpps	4.84 Mpps
64 B packet rate	0.48 Mpps	11.6 Mpps
NIC IMIX p99 latency	58 µs	0.59 µs
NIC p99, ML 1:7	241 µs	4.2 µs

Resource note

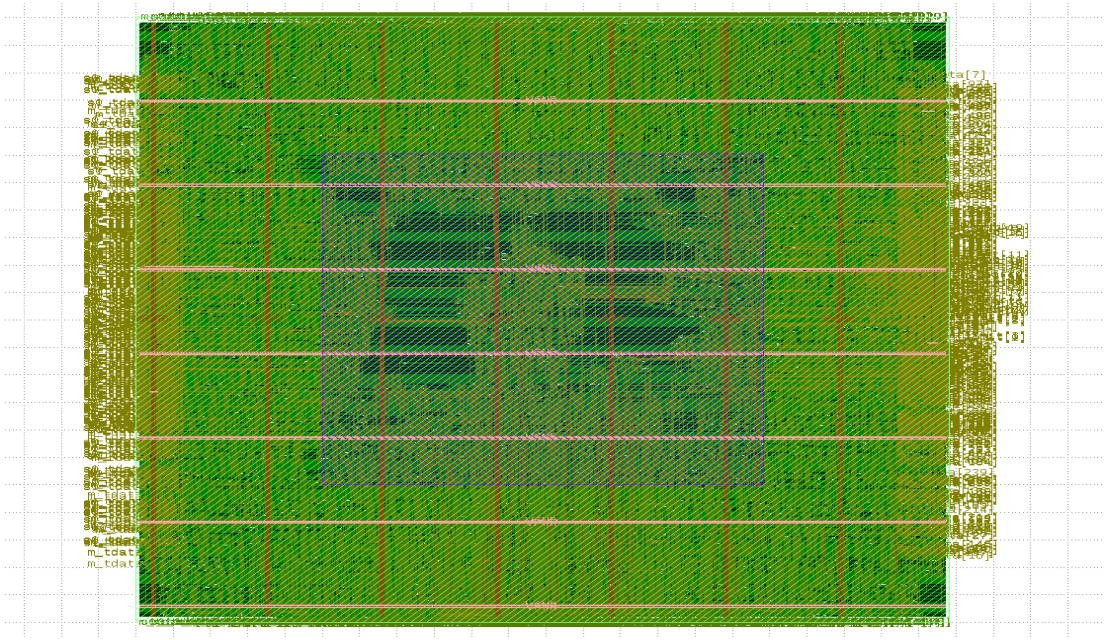
BRAM / URAM / DSP = 0. Separate supplied synthesis report: 57,184 LUTs, 39,225 FFs; tool power estimate 3.499 W (medium confidence).

SKY130 — physical implementation status

Reported status on 24 October 2026; interconnect GDSII completed, full-bridge physical sign-off not confirmed.



Layout view 1 · dense cell/routing display



Layout view 2 · metal/routing display

Interconnect (completed)

1.20 mm² die · 41% utilization · ~49k cells · 104 mW

DRC/LVS/antenna clean; hold met. Slow-corner setup −0.95 ns.

Full bridge (open)

≈3.9 mm² standard-cell area at narrow-profile synthesis.

Routing/sign-off not yet demonstrated. X.1 later improved interconnect setup to −0.52 ns without new GDS stream-out.

Why full-bridge GDSII was not completed by submission

The limiting factors were physical-implementation complexity, available compute and the submission window.

1. RTL and performance target

Wide 512-bit processing, parallel CRC/FEC, replay buffering and pipelining supported high simulation throughput with error-free delivery in the tested cases. Such a datapath also makes ASIC synthesis, placement and routing more demanding.

2. ASIC flow and compute

The SKY130 flow exposed Yosys latch/logic-loop issues in the RX correction path and long full-design runs. We made X.1–X.4 ASIC-compatibility fixes and introduced a narrow, parameterized implementation profile.

3. Timing of the final runs

Our initial computing resources were insufficient to complete the full physical-design runs within the available time. Higher-capacity campus compute was identified and secured, but debugging and reruns extended into the submission deadline.

Evidence at cutoff

Interconnect GDSII generated and DRC/LVS/antenna checks clean; full bridge synthesized (~3.9 mm² of standard cells), but full-bridge routing and sign-off remained open.

Latest team-reported RTL simulation: 6.71 GB/s with zero observed data mismatches; earlier documented result: 6.17 GB/s. Final FPGA timing: 255 MHz and positive WNS (+0.072 ns).

Engineering decisions beyond a basic RTL bridge

What we actually implemented and tested—not claims of industry-first functionality.

1. Two workloads on one fabric

Programmable weighted arbitration shares the link between SmartNIC traffic and ML tensor streaming, with independent channel sessions.

2. Actual application-level profiling

An in-house analyzer tested IMIX traffic, ML tensors, mixed weights, 4 KB crossings, packet rate and latency—not just Fmax.

3. Design for recoverability

ACK/NAK replay, CRC/FEC correction, link-drop/retrain cases, resets, overflow and error injection were exercised.

4. Regression protection

46 guard checks and a 40-seed randomized suite helped prevent repaired cross-layer bugs from reappearing during optimization.

5. Measured optimization trade-offs

We documented the 118 → 49 MHz timing regression, traced its CRC-64 critical path and recovered to 255 MHz.

6. Two technology profiles

Wide FPGA configuration and resource-reduced SKY130 configuration share RTL, with Yosys-safe logic for ASIC experiments.

Remaining engineering work Full-bridge SKY130 sign-off, complete I/O timing constraints and improved heavy-noise throughput.